

Problem Description & Metrics

Problem:

- Need to quantitatively analyze morphological changes in a specific catalyst interlayer of PEM electrolyzer cross-sections imaged via cryo-STEM.
- Focus on understanding degradation effects by tracking changes across multiple images.
- Challenge: Isolate the target catalyst layer (often a mid-intensity, porous band) from adjacent layers (membrane, other catalyst/support layers) and migrated particles.
- Goal: Develop a robust, automated method using computer vision to extract meaningful metrics.

Key Metrics Calculated:

- **Thickness:**
 - Mean, Median, Min, Max, Standard Deviation (μm or pixels)
 - Coefficient of Variation (CV %) - Normalized thickness variability
 - Percentage Near Zero Thickness (%) - Quantifies thinning/gaps
- **Roughness/Shape:**
 - RMS Roughness (Top & Bottom Boundaries, μm or pixels)
 - Contour Length (Top & Bottom Boundaries, μm or pixels)
 - Tortuosity (Top & Bottom Boundaries, dimensionless) - Path length / Projected length
- **Internal Structure/Texture:**
 - Internal Intensity (Mean, Std Dev, Skewness, Kurtosis) - Based on pixel values within the mask
 - Porosity Area Fraction (%) - Estimate based on intensity thresholding within the mask

Analysis Approach & Methodology

Automated Image Analysis Pipeline:

1. Load & Scale:

1. Load the original cryo-STEM image (grayscale).
2. Calculate pixel size ($\mu\text{m}/\text{pixel}$) using either:
 1. Horizontal Field Width (HFW) from image metadata (Default).
 2. Or manual input based on scale bar measurement.

2. Define Region of Interest (ROI):

1. Crop the image using user-defined pixel coordinates (Y-start, Y-end, X-start, X-end) to focus analysis and exclude irrelevant regions (e.g., other layers, scale bar).

3. Pre-processing (on Cropped Image):

1. Apply Gaussian Blur to reduce noise.
2. Apply Contrast Limited Adaptive Histogram Equalization (CLAHE) to enhance local contrast.

4. Segmentation (Intensity + Filtering):

1. Apply intensity range thresholding (using lower/upper percentiles) to select pixels within the target layer's typical brightness range.
2. Remove small noise objects below a minimum size threshold.
3. Identify all remaining connected components.

4. Filter Components: Keep only components that meet specific criteria:

1. *Vertical Position:* Centroid falls within a defined fraction of the

cropped image height (avoids top/bottom edges).

2. *Aspect Ratio:* Width-to-height ratio exceeds a minimum threshold (favors horizontal layers).

5. (Optional) Apply morphological closing to bridge small gaps in the accepted layer components.

5. Boundary Extraction:

1. Find the uppermost and lowermost pixels for each column within the final segmented mask.

6. Metric Calculation:

1. Compute thickness, roughness, tortuosity, internal intensity, and porosity metrics based on the extracted boundaries and the pixel values within the final mask.

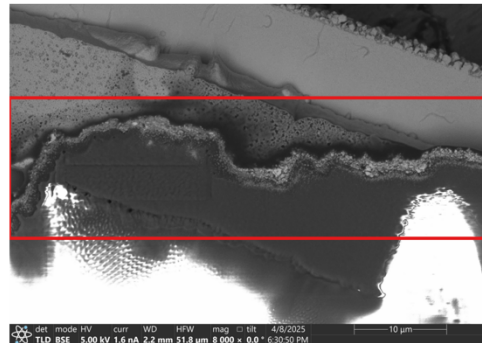
7. Visualization & Output:

1. Generate a multi-panel figure showing:
 1. Original image with crop box.
 2. Cropped image with segmentation overlay.
 3. Final mask with boundaries highlighted.
 4. (Optional) Porosity visualization.
 5. Smoothed thickness profile plot.
 6. Thickness distribution histogram.
2. (Optional) Save the figure (e.g., TIFF) and calculated metrics (e.g., TXT) to files.

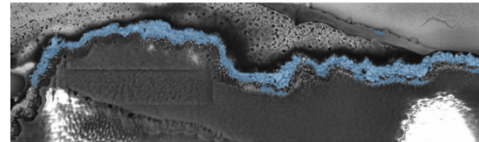
Analysis Results

PEM Catalyst Layer Analysis Results

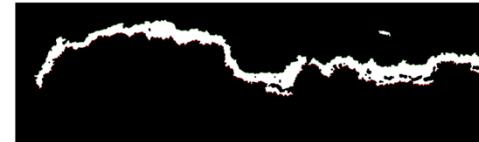
Original Image & Crop Area



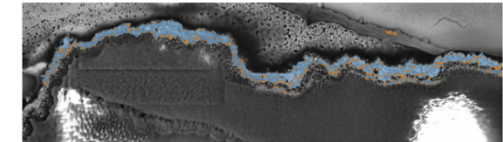
Cropped Image + Overlay



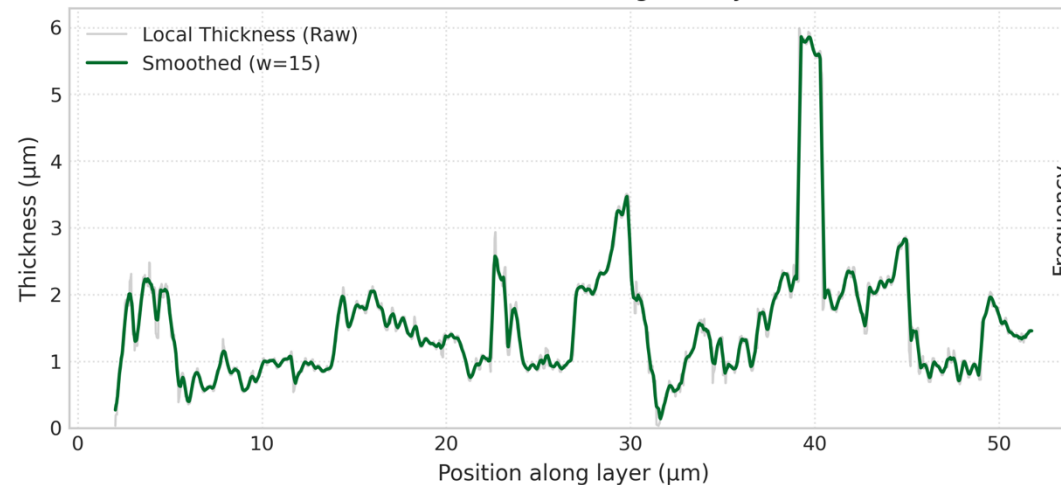
Segmented Layer + Boundaries



Detected Pores - 14.2% Area



Thickness Profile along the Layer



Thickness Distribution Histogram

